



OPEN Prognostic relevance of TROP2 expression in patients with non-small cell lung cancer receiving immunotherapy

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Trophoblast surface antigen 2 (TROP2) is strongly expressed in patients with non-small cell lung cancer (NSCLC), and its overexpression is closely associated with a worse prognosis. Recently, a novel TROP2-directed antibody–drug conjugate (ADC) has been developed, and TROP2 has been identified as a therapeutic target for NSCLC. In this study, we investigated whether TROP2 expression can predict the outcome after combined immunotherapy with programmed death-1 (PD-1) blockade plus cytotoxic T-lymphocyte antigen-4 (CTLA4) antibody. The present study included 110 patients with advanced NSCLC who received nivolumab plus ipilimumab (Nivo-Ipi) between 2020 and 2022 at our institution. TROP2 and Ki-67 protein expression were evaluated by immunohistochemistry. Inflammatory and nutritional indices were investigated with several variables, including neutrophil to lymphocyte ratio (NLR), platelet to lymphocyte ratio (PLR), systemic immune-inflammation index (SII), prognostic nutritional index (PNI), advanced lung cancer inflammation index (ALI), and Glasgow prognostic score. TROP2 was highly expressed in 46 (41.8%) patients, and its overexpression was significantly associated with poor response to Nivo-Ipi therapy. Univariate analysis of all patients identified performance status, liver metastases, bone metastases, NLR, TROP2, PLR, SII, PNI, and ALI as significant predictors of progression-free survival (PFS) and overall survival (OS) after Nivo-Ipi treatment. Multivariate analysis identified TROP2 overexpression as an independent prognostic predictor for PFS and OS. Specifically, TROP2 overexpression was significantly associated with shorter PFS and OS in the subgroups of PD-L1 < 1% or non-adenocarcinoma histology. TROP2 expression could be a significant predictor for outcome after Nivo-Ipi treatment in patients with advanced NSCLC.

Keywords TROP2, PD-1, CTLA4, Lung cancer, Resistance, Prognosis

Immunotherapy such as that using immune checkpoint inhibitors (ICIs) is generally administered to patients with advanced non-small cell lung cancer (NSCLC). Immunotherapy consists of programmed death-1 (PD-1) blockade or anti-PD-1 antibody plus cytotoxic T-lymphocyte antigen-4 (CTLA4) antibody determined by the expression of programmed death-ligand-1 (PD-L1) in cancer cells. Currently, there is no optimal biomarker for predicting the efficacy of anti-PD-1 (nivolumab) plus anti-CTLA4 (ipilimumab; Nivo-Ipi) antibody-based immunotherapy for the treatment of NSCLC.

Recently, several new agents targeting tumor specific molecules have been developed to establish late-line standard of care for patients with advanced NSCLC. Of these, trophoblast surface antigen 2 (TROP2) is a transmembrane glycoprotein expressed in NSCLC, with its high expression closely associated with a poor outcome, making it a promising therapeutic target¹. Recent prospective studies demonstrated promising antitumor activity of datopotamab deruxtecan (Dato-DXd), a novel TROP2-directed antibody–drug conjugate (ADC), in pretreated patients with advanced NSCLC². Dato-DXd significantly improved progression-free

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survival (PFS) compared to docetaxel in a phase III study³. However, it is unclear whether TROP2 expression can be used as a biomarker of therapeutic efficacy and outcome in patients with advanced NSCLC.

Previous studies have reported that TROP2 overexpression in tumor cells is a predictor of worse prognosis in patients with NSCLC harboring early-stage disease^{4,5}. Further, Jiang et al. showed that TROP2 overexpression was closely correlated with poor prognosis in patients with advanced NSCLC who received platinum-based chemotherapy or chemo-radiotherapy⁶. Currently, a randomized phase III study comparing Dato-DXd with pembrolizumab to pembrolizumab alone is ongoing worldwide in patients with advanced NSCLC⁷. A recent study observed a significant association of TROP2 overexpression with a poor outcome on PD-1 blockade based on transcriptomic data from 891 patients with NSCLC in two large, randomized trials⁸. A preclinical study also suggested that TROP2 expression could affect the functions of cytotoxic T cells by leading them towards apoptosis⁹. However, it remains unclear whether the TROP2 expression can be used as a biomarker for the efficacy of treatment with CTLA4 antibody in addition to PD-1 blockade in patients with advanced NSCLC.

Here, we conducted a retrospective study to evaluate the clinicopathological relevance of TROP2 expression in patients with advanced NSCLC receiving Nivo-Ipi therapy.

Results

Patient demographics

Patient characteristics are summarized in Table 1. The median age was 72 years, ranging from 45 to 86 years. The detailed information was as follows: 89 patients were male and 21 were female; 37 patients had a PS of 0 and 50 had a PS of 1; 56 patients had adenocarcinoma (AD), 41 had squamous cell carcinoma (SCC), and 13 had tumors of other histological types; 47 patients had PD-L1 < 1%, 51 had 1–49%, 10 had ≥ 50%, and 2 patients had an unknown status. The objective response rate (ORR) of 110 patients was 37.3% [1 complete response (CR), 40 partial response (PR), 33 stable disease (SD), 28 progressive disease (PD), and 8 not evaluated (NE)].

Nivo-Ipi therapy was mainly chosen for patients with PS 0–1 and low PD-L1 expression (particularly ≤ 10%). It was occasionally used in patients with PS 2–3, particularly in younger patients or those with low PD-L1 expression. A few patients with PD-L1 ≥ 50% also received this therapy, usually when chemotherapy was considered inappropriate or when a durable immune response was anticipated. Driver gene mutation status was available in 98 patients. Among them, 89 had no mutations, while nine patients harbored driver mutations (*EGFR*, n = 1; *KRAS*, n = 7; others, n = 2). No *ALK* rearrangements were detected. Mutation status was unavailable in 12 patients.

Immunohistochemical evaluation

TROP2 protein expression was immunohistochemically detected in 110 tumors. Of them, 86 were biopsy samples and 24 were surgically resected. Figure 1 displays representative IHC images. TROP2 immunostaining was detected in the cytoplasm and the plasma membrane. Overexpression of TROP2 was observed in 46 (41.8%) patients (AD: 20, SCC: 20, Others: 6). Cytoplasmic staining of TROP2 with an intensity score of 3 was observed in 64 (58.1%) patients (AD: 29, SCC: 26, Others: 9). TROP2 staining with a proportion score of 4 was observed

Baseline characteristics (n = 110)	
Age	
< 75/≥ 75 years	72 (65.5%)/38 (34.6%)
ECOG performance status	
0/1/2/3	37 (33.6%)/50 (45.5%)/16 (14.5%)/7 (6.4%)
Sex	
Male/Female	89 (80.9%)/21 (19.1%)
Smoking index	
Yes/No	92 (83.6%)/18 (16.4%)
Staging status	
III/IV/Rec. after surgery/Rec. after RT	6 (5.4%)/68 (61.8%)/24 (21.8%)/12 (10.8%)
Histology	
AD/SCC/Others	56 (50.9%)/41 (37.3%)/13 (11.8%)
PD-L1 (TPS) (%)	
≥ 50/1–49% / < 1%/Unknown	10 (9.1%)/51 (46.4%)/47 (42.7%)/2 (1.8%)
Any grade irAEs	
Present/Absent	77 (70.0%)/33 (30.0%)
Response by RECIST	
CR/PR/SD/PD/NE	1 (0.9%)/40 (36.4%)/33 (30.0%)/28 (25.4%)/8 (7.3%)

Table 1. Patient characteristics. Abbreviations: ECOG, eastern cooperative oncology group; Rec. recurrence; RT, radiation; AD, adenocarcinoma; SCC, squamous cell carcinoma; PD-L1, programmed death ligand-1; TPS, tumor proportional score; irAEs, immune-related adverse events; RECIST, response evaluation criteria in solid tumors; CR, complete response; PR, partial response; SD, stable disease; PD, progressive disease; NE, not evaluable.

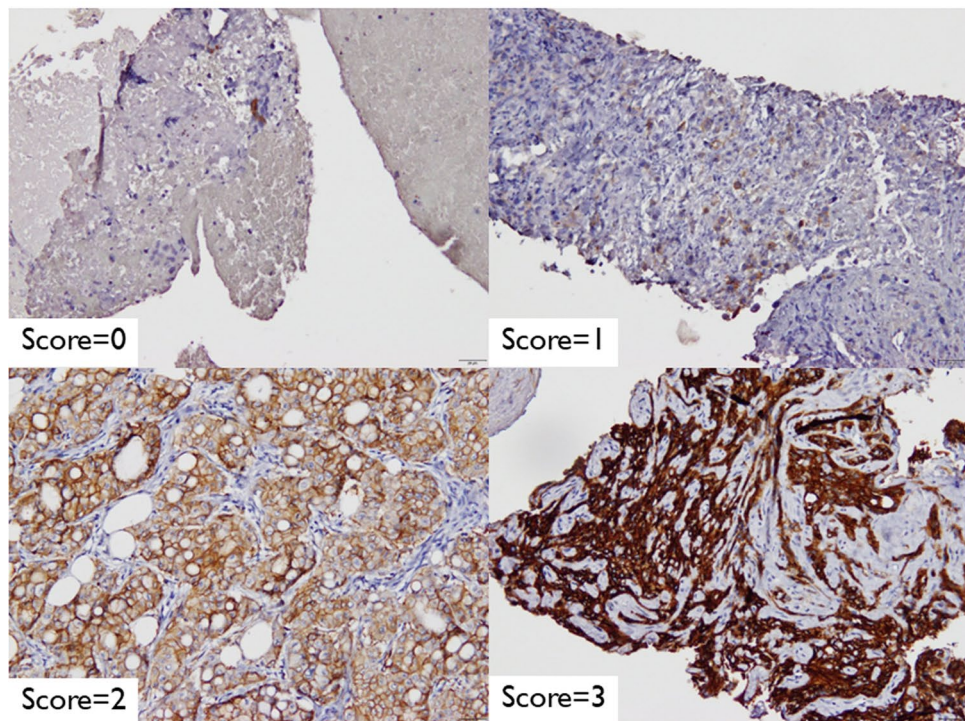
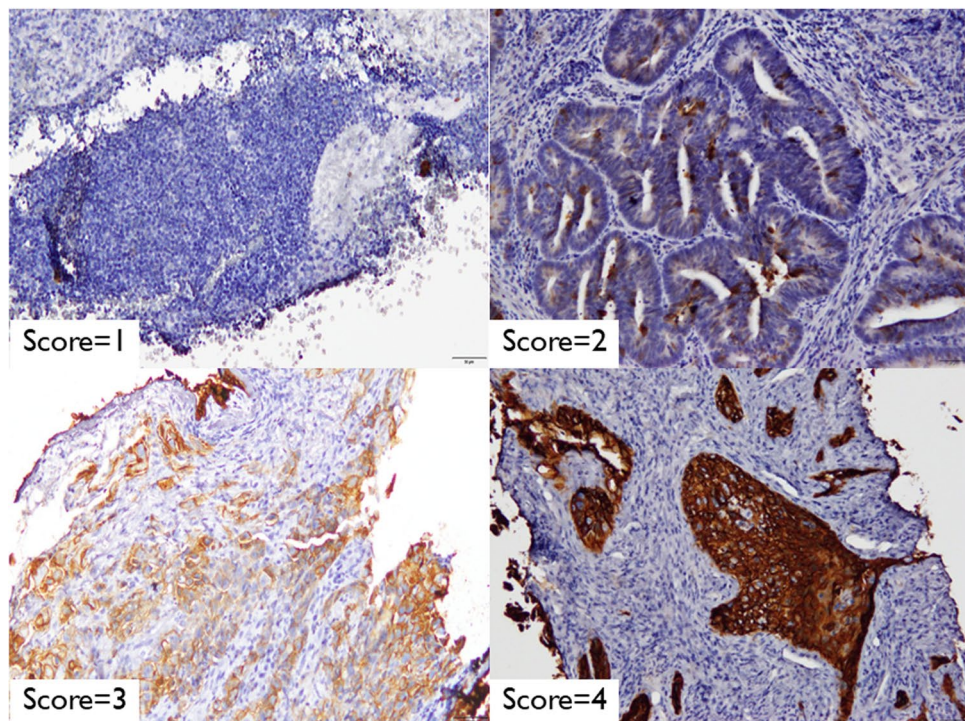
A. Intensity score (0 – 3)**B. Proportion score (1 – 4)**

Fig. 1. Representative immunohistochemical staining of TROP2 expression in lung cancer. The assessment of TROP2 expression was performed based on intensity score of 0 to 3 (A) and proportion score of 1 to 4 (B).

in 68 (61.8%) patients (AD: 36, SCC: 25, Others: 7). The patient characteristics according to TROP2 expression are shown in Table 2. Overexpression of TROP2 was significantly associated with poor response.

TROP2 expression with an intensity score of 3 was significantly related to lower incidence of brain metastasis and low leukocyte count. TROP2 expression with a proportion score of 4 was also significantly related to the

Different variables		All pats n = 110 (%)	TROP2 (Int. × Prop.)			TROP2 (Prop.)			TROP2 (Int.)		
			< 12 n = 64 (58.2%)	12 n = 46 (41.8%)	p-value	< 4 n = 42 (38.2%)	4 n = 68 (61.8%)	p-value	< 3 n = 46 (41.8%)	3 n = 64 (58.2%)	p-value
Age	< 75	72 (65.5)	46 (71.9)	26 (56.5)	0.095	29 (69.1)	43 (63.2)	0.533	33 (71.7)	39 (60.9)	0.240
	≥ 75 yrs	38 (34.6)	18 (28.1)	20 (43.4)		13 (31.0)	25 (36.8)		13 (28.3)	25 (39.1)	
Sex	Male	89 (80.9)	54 (84.4)	35 (76.1)	0.275	35 (83.3)	54 (79.4)	0.611	39 (84.8)	50 (78.1)	0.381
	Female	21 (19.1)	10 (15.6)	11 (23.9)		7 (16.7)	14 (20.6)		7 (15.2)	14 (21.9)	
ECOG PS	0–1	87 (79.1)	53 (82.8)	34 (73.9)	0.260	34 (81.0)	53 (77.9)	0.706	39 (64.8)	48 (75.0)	0.213
	2–3	23 (20.9)	11 (17.2)	12 (26.1)		8 (19.1)	15 (22.1)		7 (15.2)	16 (25.0)	
Smoking Index	Yes	92 (83.6)	54 (84.4)	38 (82.6)	0.805	5 (11.9)	13 (19.1)	0.321	7 (15.2)	11 (17.2)	0.783
	No	18 (16.4)	10 (15.6)	8 (17.4)		37 (88.1)	55 (80.9)		39 (84.8)	53 (82.8)	
Histology	AD	56 (50.9)	36 (56.3)	20 (43.5)	0.186	20 (47.6)	36 (52.9)	0.588	27 (58.7)	29 (45.3)	0.166
	Non-AD	54 (49.1)	28 (43.8)	26 (56.5)		22 (52.4)	32 (47.1)		19 (41.3)	35 (54.7)	
Liver meta	Yes	15 (13.6)	9 (14.1)	6 (13.0)	0.877	6 (14.3)	9 (13.2)	0.876	6 (13.0)	9 (14.1)	0.878
	No	95 (83.4)	55 (85.9)	40 (87.0)		36 (85.7)	59 (86.8)		40 (87.0)	55 (85.9)	
Brain meta	Yes	19 (17.3)	15 (23.4)	4 (8.7)	0.044	5 (11.9)	14 (20.6)	0.242	12 (26.1)	7 (10.9)	0.038
	No	91 (82.7)	49 (76.6)	42 (91.3)		37 (88.1)	54 (79.4)		34 (73.9)	57 (89.1)	
Bone meta	Yes	31 (28.2)	18 (28.1)	13 (28.7)	0.988	9 (21.4)	22 (33.4)	0.216	14 (30.4)	17 (26.6)	0.656
	No	79 (71.8)	46 (71.9)	33 (71.7)		33 (78.6)	46 (67.7)		32 (69.6)	47 (73.4)	
Response	PR	41 (37.3)	29 (45.3)	12 (26.1)	0.040	19 (45.2)	22 (32.4)	0.175	21 (45.7)	20 (31.3)	0.123
	Non-PR	69 (62.7)	35 (54.7)	34 (73.9)		23 (54.8)	46 (67.7)		25 (54.4)	44 (68.8)	
PD-L1(%)	≥ 1	61 (55.5)	36 (57.1)	25 (55.6)	0.870	21 (51.2)	40 (59.7)	0.388	27 (60.0)	34 (54.0)	0.533
	< 1	47 (44.5)	27 (42.9)	20 (44.4)		20 (48.8)	27 (40.3)		18 (40.0)	29 (46.0)	
Any irAEs	Yes	77 (70.0)	47 (73.4)	30 (65.2)	0.353	34 (81.0)	43 (63.2)	0.049	33 (71.7)	44 (68.8)	0.736
	No	33 (30.0)	17 (26.6)	16 (34.8)		8 (19.1)	25 (37.7)		13 (28.3)	20 (31.3)	
G3/4 irAEs	Yes	37 (33.6)	21 (32.8)	16 (34.8)	0.829	14 (33.3)	23 (33.8)	0.958	16 (34.8)	21 (32.8)	0.829
	No	73 (66.4)	43 (67.2)	30 (65.2)		28 (66.7)	45 (66.2)		30 (65.2)	43 (67.2)	
NLR	High	52 (47.3)	30 (46.9)	22 (47.8)	0.922	21 (50.0)	31 (45.6)	0.653	24 (52.2)	28 (47.8)	0.383
	Low	58 (52.7)	34 (53.1)	24 (52.2)		21 (50.0)	37 (54.4)		22 (47.8)	36 (56.3)	
PLR	High	51 (46.4)	29 (45.3)	22 (47.8)	0.794	22 (52.4)	29 (42.6)	0.320	20 (43.5)	31 (48.4)	0.607
	Low	59 (53.6)	35 (54.7)	24 (52.2)		20 (47.6)	39 (57.4)		26 (56.5)	33 (51.6)	
SII	High	57 (51.8)	32 (50.0)	25 (54.4)	0.653	22 (52.4)	35 (51.5)	0.926	24 (52.2)	33 (51.6)	0.950
	Low	53 (48.2)	32 (50.0)	21 (45.7)		20 (47.6)	33 (48.5)		22 (47.8)	31 (48.4)	
PNI	High	59 (53.6)	36 (56.3)	23 (50.0)	0.517	20 (47.6)	39 (57.4)	0.320	25 (54.4)	34 (53.1)	0.899
	Low	51 (46.4)	28 (43.7)	23 (50.0)		22 (52.4)	29 (42.6)		21 (45.6)	30 (46.9)	
ALI	High	57 (51.8)	34 (53.1)	23 (50.0)	0.746	21 (50.0)	36 (52.9)	0.764	22 (47.8)	35 (54.7)	0.478
	Low	53 (48.2)	30 (46.9)	23 (50.0)		21 (50.0)	32 (47.1)		24 (52.2)	29 (45.3)	
GPS	High	44 (40.0)	23 (35.9)	21 (45.7)	0.305	14 (33.3)	30 (44.1)	0.262	16 (34.8)	28 (43.8)	0.344
	Low	66 (60.0)	41 (64.1)	25 (54.4)		28 (66.7)	38 (55.9)		30 (65.2)	36 (56.3)	
Leukocyte	High	59 (53.6)	37 (57.8)	22 (47.8)	0.300	23 (54.8)	36 (52.9)	0.852	30 (65.2)	29 (45.3)	0.039
	Low	51 (46.4)	27 (42.2)	24 (52.2)		19 (45.2)	32 (47.1)		16 (34.8)	35 (56.7)	
Neutrophil	High	59 (53.6)	36 (56.3)	23 (50.0)	0.517	23 (54.8)	36 (52.9)	0.852	29 (63.0)	30 (47.0)	0.094
	Low	51 (46.4)	28 (43.8)	23 (50.0)		19 (45.2)	32 (47.1)		17 (37.0)	34 (53.1)	
Lymphocyte	High	57 (51.8)	36 (56.3)	21 (45.6)	0.273	21 (50.0)	36 (52.9)	0.764	27 (58.7)	30 (46.9)	0.221
	Low	53 (48.2)	28 (43.8)	25 (54.4)		21 (50.0)	32 (47.1)		19 (41.3)	34 (53.1)	
Albumin	High	58 (52.7)	33 (51.6)	25 (54.4)	0.714	19 (45.2)	39 (57.4)	0.216	22 (47.8)	36 (56.3)	0.383
	Low	52 (47.3)	31 (48.4)	21 (45.6)		23 (54.7)	29 (42.6)		24 (52.2)	28 (43.8)	
CRP	High	54 (49.1)	32 (50.0)	22 (47.8)	0.822	25 (59.5)	29 (42.6)	0.085	24 (52.2)	30 (46.9)	0.583
	Low	56 (50.9)	32 (50.0)	24 (52.2)		17 (40.5)	39 (57.4)		22 (47.8)	34 (53.1)	
Ki-67	High	54 (49.1)	29 (46.8)	25 (54.4)	0.436	23 (54.8)	31 (47.0)	0.430	19 (43.2)	35 (54.7)	0.240
	Low	56 (50.8)	33 (52.2)	21 (45.6)		19 (45.2)	35 (53.0)		25 (56.8)	29 (45.3)	

Table 2. Patient's characteristics according to TROP2 expression. Cut-off values: NLR, 4.13; PLR, 240.6; SII, 134.1; PNI, 42.6; ALI, 18.1; Leukocyte, 7250; Neutrophil, 5302; Lymphocyte, 1223; Albumin, 3.6; CRP, 0.84; Ki-67 28.9. GPS was categorized as low (0) or high (1–2). AD, adenocarcinoma; ALI, advanced lung cancer inflammation index; CRP, C-reactive protein; ECOG, eastern cooperative oncology group; GPS, Glasgow prognostic score; Int., intensity; irAEs, immune-related adverse events; meta, metastasis; NLR, neutrophil to lymphocyte ratio; PD-L1, programmed death ligand-1; PLR, platelet to lymphocyte ratio; PNI, prognostic nutrition index; PR, partial response; Prop., proportion; PS, performance status; SII, systemic immune inflammation index; TROP2, Trophoblast surface antigen 2; yrs, years.

frequency of immune-related Adverse Events (irAEs). Patient characteristics according to PD-L1, histology, and TROP2 expression are shown in Table A1 (Supplementary Information).

Survival analysis

The median PFS and OS were 210 and 413 days, respectively. Eighty-one patients experienced tumor recurrence, and 59 died due to disease progression. The Kaplan–Meier curves for PFS and OS according to TROP2 expression are depicted in Fig. 2. Overexpression of TROP2 was associated with a shorter PFS and OS in all patients, and worse prognosis in PFS and OS was observed in patients with TROP2 overexpression even after stratification by PD-L1 expression rates (PD-L1 < 1%; Fig. 2C,D, PD-L1 ≥ 1%; Fig. 2E,F) and histological type (AD; Fig. 2G,H, non-AD; Fig. 2I,J). Based on these subgroups, patients with TROP2 overexpression had significantly shorter PFS and OS in PD-L1 < 1% or non-AD histology groups.

Next, we conducted a survival analysis based on the intensity and proportional expression of TROP2, including subgroups defined by PD-L1 expression status (PD-L1 < 1% and PD-L1 ≥ 1%) and histology (AD and non-AD) (Fig. 3). High intensity of TROP2 expression was associated with worse PFS and OS in all patients, whereas high proportional expression of TROP2 showed no prognostic value. Stratified by PD-L1 expression status, high intensity of TROP2 expression was significantly associated with worse OS, while no significant association with OS was observed with high proportional TROP2 expression. In patients with AD, high intensity of TROP2 expression was closely associated with worse PFS. Conversely, in patients with non-AD histology, high proportional expression of TROP2 was significantly associated with worse PFS.

The results of univariate and multivariate analyses for PFS and OS in all patients are summarized in Table 3. Univariate analysis of all patients identified PS, liver metastases, bone metastases, NLR, TROP2, PLR, SII, PNI, and ALI as significant predictors of PFS and OS. The univariate log-rank test enabled screening for variables with $p < 0.05$ for subsequent multivariate analysis. The results of the multivariate analysis identified liver metastases, bone metastases, TROP2, and SII as independent predictors of PFS. Liver metastases, bone metastases, NLR, TROP2, SII, and PNI were also identified as independent predictors of OS.

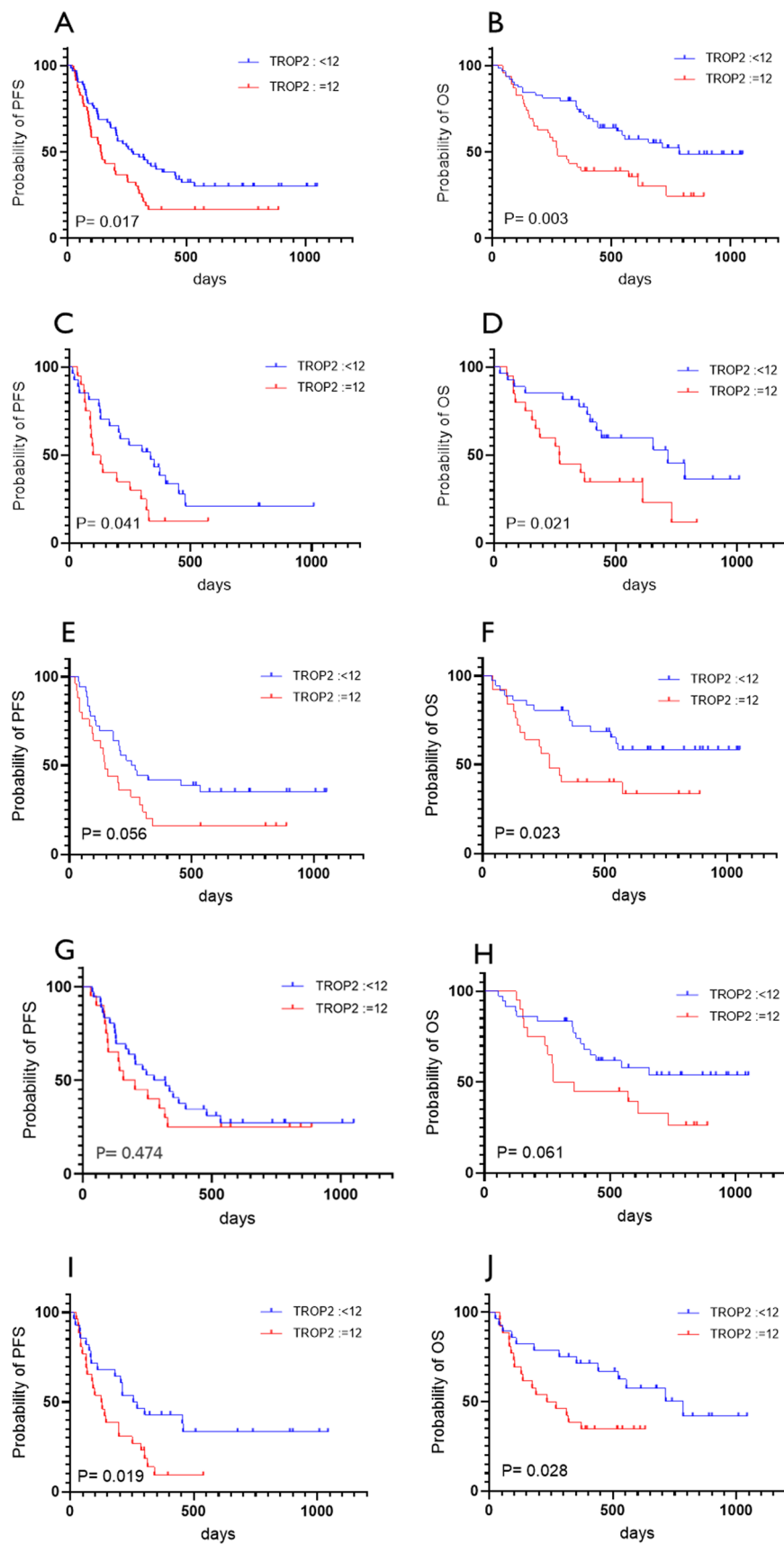
Discussion

Here, we conducted a clinicopathological study to evaluate the prognostic significance of TROP2 expression in patients with advanced NSCLC who received Nivo-Ipi therapy. We found that overexpression of TROP2 was a significant biomarker for predicting worse outcome after Nivo-Ipi, regardless of PD-L1 expression. Apart from TROP2 expression, liver and bone metastases and inflammatory markers such as NLR and SII were also identified as significant predictors. Our findings suggest that the expression of TROP2 was not associated with liver and bone metastases, and inflammatory and nutritional induces. Therefore, TROP2 may be a unique predictor for the therapeutic efficacy of Nivo-Ipi treatment in patients with NSCLC. In particular, TROP2 expression could be a helpful predictor for patients with PD-L1 < 1% and non-AD histology.

The expression of TROP2 was immunohistochemically specific for tumor cells, and correlated with poor outcomes in patients with NSCLC^{4–6,8}. TROP2 promotes tumor cell growth, proliferation, and metastasis by regulating the calcium signaling pathway and cyclin expression¹⁰. TROP2 overexpression has been reported in patients with breast, pancreatic, prostate, and ovarian cancers, as well as NSCLC¹⁰. Kobayashi K et al. reported that TROP2 overexpression was observed in 87 out of 130 lung adenocarcinoma cases (66%)⁵. TROP2 overexpression was identified in 41.8% of cases in the present study, consistent with the results of previous reports indicating TROP2 overexpression in NSCLC^{4–6}. In this study, high TROP2 expression was significantly associated with lower incidence of brain metastasis. Previous studies have suggested that SCC shows a lower frequency of brain metastasis and numerically higher rates of TROP2 overexpression compared with AD^{11,12}. In our cohort, SCC accounted for 41 out of 110 cases (37.3%), and TROP2 overexpression was numerically more frequent in SCC (48.7%) compared with AD (35.7%). According to a population-based study using the SEER–NPCR database, SCC constituted 29.4% of NSCLC cases in the United States¹³, indicating that the proportion of SCC in our study was relatively higher than in previous reports. Another possible explanation is that patients with high TROP2 expression may experience rapid disease progression, leading to death before developing brain metastases. Taken together, the histological composition of our cohort and the potential for rapid disease progression in TROP2-high tumors may underlie the lower incidence of brain metastases observed in these patients.

Recent studies revealed a close association between TROP2 expression and the tumor immune microenvironment, and that TROP2 overexpression resulted in reduced levels of T-cell infiltration in patients with NSCLC^{8,14}. Bessede et al. described that TROP2 overexpression was closely related to therapeutic resistance to ICIs but not to chemotherapy in patients with advanced NSCLC⁸. We did not identify a significant correlation between TROP2 and PD-L1 expression, which has not been investigated in detail in previous studies. The absence of a correlation between TROP2 and PD-L1, despite TROP2's involvement in the immune microenvironment such as in tumor-infiltrating lymphocytes (TILs), indicates that TROP2 may have a distinct and independent role in tumor immunity. Notably, the intracellular, but not the membrane, expression of TROP2 was significantly correlated with a worse outcome after PD-1 blockade monotherapy⁸. The functional protein domain of TROP2 is released from the membrane and accumulates within the nucleus of tumor cells⁸. We observed that intense TROP2 expression was closely associated with a poor outcome, whereas the proportional of TROP2 expression was not a significant biomarker for the prediction of Nivo-Ipi efficacy. As our study defined the intensity level as significant staining of intracellular and membrane lesions, an intensity score of 3 may be due to the intracellular expression of TROP2. Therefore, not only the membrane, but also intracellular expression of TROP2 could predict the resistance to Nivo-Ipi therapy in patients with advanced NSCLC.

TROP2 was associated with decreased OS in some solid tumors, affecting the tumor immune microenvironment¹⁴. The expression of TROP2 in cervical cancer was associated with increased intratumoral



◀ **Fig. 2.** Kaplan–Meier survival curve according to TROP2 expression in all patients [PFS (A) and OS (B)], subgroup of PD-L1 < 1% [PFS (C) and OS (D)], those of PD-L1 ≥ 1% [PFS (E) and OS (F)], those of AD [PFS (G) and OS (H)], and those of non-AD [PFS (I) and OS (J)]. Median PFS and OS values, hazard ratios (HR) with 95% confidence intervals (CI), and log-rank p-values were as follows: all patients: median PFS 273 vs 140 days (HR = 1.30, 95% CI 1.05–1.63, $p = 0.017$) [A]; median OS 784 vs 272 days (HR = 1.46, 95% CI 1.13–1.89, $p = 0.003$) [B]. PD-L1 < 1%: median PFS 336 vs 113 days (HR = 1.41, 95% CI 1.01–1.98, $p = 0.041$) [C]; median OS 714 vs 269 days (HR = 1.54, 95% CI 1.06–2.25, $p = 0.021$) [D]. PD-L1 ≥ 1%: median PFS 262 vs 145 days (HR = 1.33, 95% CI 0.99–1.79, $p = 0.056$) [E]; median OS NR vs 273 days (HR = 1.50, 95% CI 1.04–2.15, $p = 0.023$) [F]. AD: median PFS 298 vs 179 days (HR = 1.13, 95% CI 0.82–1.56, $p = 0.474$) [G]; median OS NR vs 314 days (HR = 1.41, 95% CI 0.98–2.03, $p = 0.061$) [H]. non-AD: median PFS 262 vs 127 days (HR = 1.45, 95% CI 1.05–1.99, $p = 0.019$) [I]; median OS 784 vs 251 days (HR = 1.52, 95% CI 1.04–2.24, $p = 0.028$) [J].

CD3+/CD8+TILs and PD-L1 expression¹⁵. Conversely, the NSCLC tumors with higher TROP2 expression had lower gene expression of immune cell markers and were linked to a negative response to ICIs^{8,14}. Moreover, patients with advanced NSCLC exhibited reduced levels of T gamma delta, Th1 cells, and macrophages in high TROP2-expressing tumors⁸. Thus, overexpression of TROP2 may be associated with reduced T-cell infiltration in the tumor microenvironment in patients with NSCLC, which could contribute to ICI resistance. Our study suggests that TROP2 overexpression was a potential biomarker of therapeutic resistance to not only PD-1 blockade but also to anti-CTLA4 antibody.

ADCs targeting TROP2 are undergoing clinical trials for various tumors including lung cancer¹⁰. For example, sacituzumab govitecan has been approved for the treatment of triple-negative breast cancer, advanced urothelial carcinoma, and HR+/HER2-breast cancer¹⁰. A prospective study of TROPION-Lung02 examined the combination of Dato-DXd and pembrolizumab in 47 patients with advanced NSCLC, and identified an objective response rate of 38% and median PFS of 10.8 months¹⁶. The efficiency of TROP2-targeted ADCs is predicted to increase with TROP2 overexpression. Previous studies in breast cancer demonstrated an association between TROP2 expression and therapeutic efficiency, although a similar correlation was also demonstrated in lung cancer¹⁷. Considering the high frequency of TROP2 overexpression observed in our study, the potential effectiveness of ADCs in patients receiving Nivo-Ipi therapy is expected. Further studies are required to elucidate clinical benefits from a combination of ICIs and anti-TROP2 agents.

There are several limitations in our study. First, our sample size was limited, which may cause some bias in the results. Second, we did not examine whether the predictive significance of TROP2 expression could be different between patients receiving ICIs and those receiving chemotherapy. A previous study showed that TROP2 overexpression was a significant predictor for PD-1 blockade monotherapy efficacy⁸. It remains unclear whether the expression of TROP2 varies according to different therapeutic regimens. Finally, the quantitative assessment of TROP2 expression within tumor cells remains unclear. The criteria used in our study was determined based on previous studies^{4–6}. However, the calculation of tumor proportion and intensity may be necessary for the accurate evaluation of TROP2 expression. Further studies are warranted to elucidate the optimal assessment of TROP2 expression using IHC. In addition, differences in treatment duration among patients may have influenced the observed incidence of irAEs, and landmark analysis was not performed in this retrospective study.

In conclusion, TROP2 expression was identified as a significant predictor for Nivo-Ipi treatment outcome in patients with advanced NSCLC. The prognostic role of TROP2 was uniform regardless of PD-L1 expression, however, TROP2 may be a unique predictor related to therapeutic efficacy in patients with PD-L1 < 1% and non-adenocarcinoma histology. Further studies, including a prospective study combining Nivo-Ipi with ADCs targeting TROP2, are required in patients with advanced NSCLC.

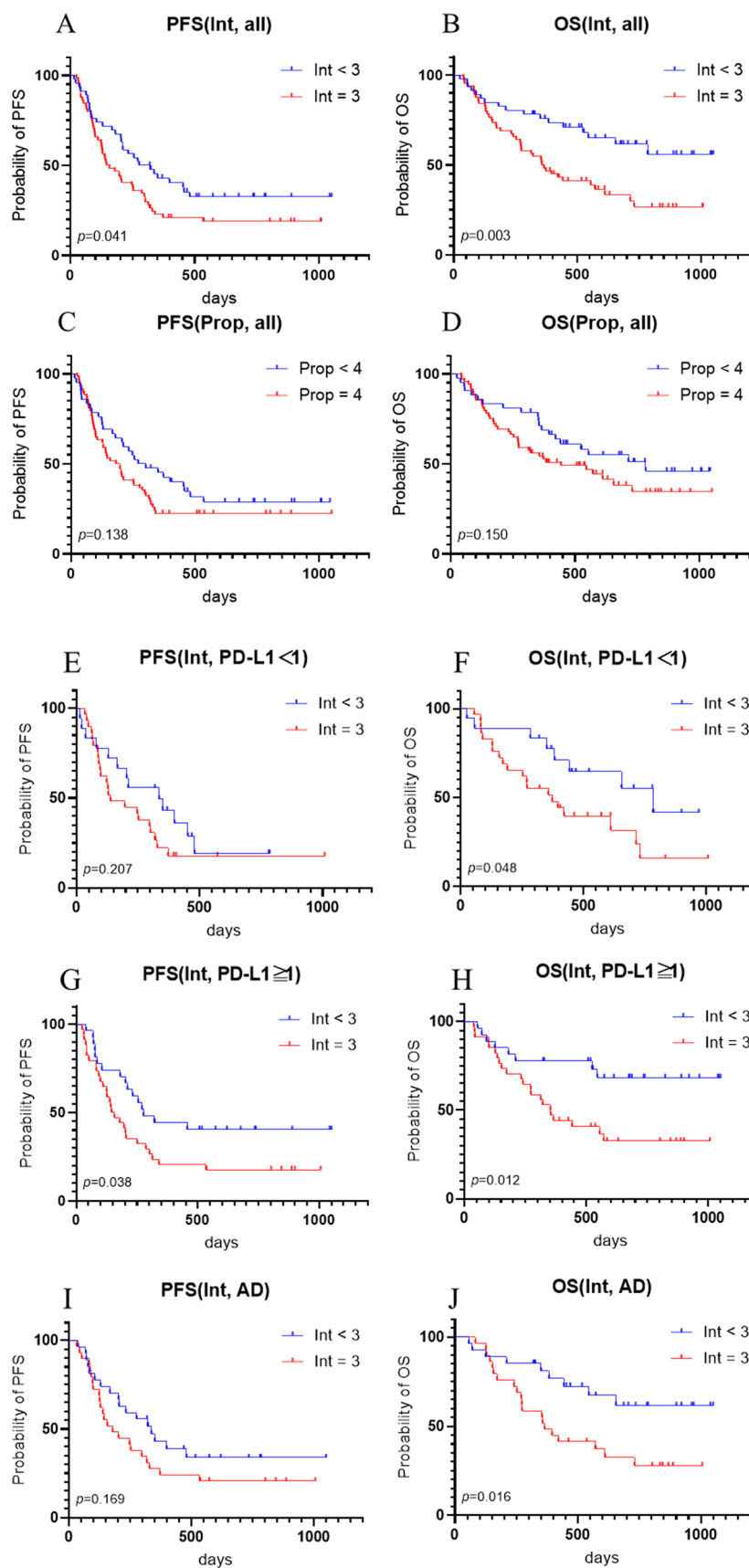
Methods

Patients

Between December 2020 and December 2022, 166 patients with NSCLC were treated with Nivo-Ipi therapy as a first-line treatment at our institution. Of these, tumor specimens at baseline were unavailable for 48 patients and immunostaining assessment on tissue slides was difficult for eight. Thus, 110 patients were eligible for this study, some of whom have been reported previously¹⁸. This study was approved by the Institutional Review Board (IRB) of Saitama Medical University, International Medical Center (approval number: 20–125). The requirement for written informed consent was waived by the IRB due to the retrospective study design. In accordance with the IRB-approved protocol, an opt-out method was employed, and patients were given the opportunity to decline participation after public disclosure of study details. All procedures involving human participants were conducted in accordance with the ethical standards of the institutional review board and the 1964 Declaration of Helsinki and its later amendments.

Treatment and evaluation

All patients were intravenously administered nivolumab (240 mg/day) once every 3 weeks and 1 mg/kg ipilimumab once every 6 weeks, as previously described¹⁹. The chief physicians evaluated the test results (physical examinations, complete blood counts, and biochemical tests) and graded the adverse events (AEs). Moreover, white blood cells, neutrophils, lymphocytes, and albumin were extracted within one week of the first cycle of Nivo-Ipi. Toxicity was graded based on the Common Terminology Criteria for AEs, version 4.0. Tumor response was assessed according to the Response Evaluation Criteria in Solid Tumors (RECIST), version 1.1²⁰. irAEs were assessed from the initiation of Nivo-Ipi therapy until discontinuation of therapy due to disease progression, death, or other reasons.



◀ **Fig. 3.** Kaplan–Meier survival curves for PFS and OS in all patients based on TROP2 intensity [PFS (A), OS (B)] and proportion [PFS (C), OS (D)]. Subgroup analyses by PD-L1 expression (PD-L1 < 1%: PFS (E), OS (F); PD-L1 ≥ 1%: PFS (G), OS (H)) and histology (AD: PFS (I), OS (J); non-AD: PFS (K), OS (L)) for TROP2 intensity. Those of PD-L1 expression (PD-L1 < 1%: PFS (M), OS (N); PD-L1 ≥ 1%: PFS (O), OS (P)) and histology (AD: PFS (Q), OS (R); non-AD: PFS (S), OS (T)) for TROP2 proportion. AD, adenocarcinoma. Median PFS and OS values, hazard ratios (HR) with 95% confidence intervals (CI), and log-rank p-values were as follows: all patients (Intensity): median PFS 298 vs 151 days (HR = 1.26, 95% CI 1.01–1.59, $p = 0.041$) [A]; median OS NR vs 356 days (HR = 1.52, 95% CI 1.14–2.02, $p = 0.003$) [B]. all patients (Proportion): median PFS 285 vs 189 days (HR = 1.19, 95% CI 0.94–1.49, $p = 0.138$) [C]; median OS 784 vs 443 days (HR = 1.22, 95% CI 0.93–1.60, $p = 0.150$) [D]. PD-L1 < 1% (Intensity): PFS 336 vs 139 days (HR = 1.25, 95% CI 0.88–1.76, $p = 0.207$) [E]; OS 784 vs 372 days (HR = 1.50, 95% CI 0.99–2.28, $p = 0.048$) [F]. PD-L1 ≥ 1% (Intensity): PFS 276 vs 151 days (HR = 1.38, 95% CI 1.01–1.88, $p = 0.038$) [G]; OS NR vs 353 days (HR = 1.65, 95% CI 1.10–2.49, $p = 0.012$) [H]. AD (Intensity): PFS 336 vs 178 days (HR = 1.24, 95% CI 0.91–1.71, $p = 0.169$) [I]; OS NR vs 365 days (HR = 1.60, 95% CI 1.08–2.37, $p = 0.016$) [J]. non-AD (Intensity): PFS 255 vs 139 days (HR = 1.23, 95% CI 0.88–1.72, $p = 0.211$) [K]; OS 784 vs 353 days (HR = 1.40, 95% CI 0.93–2.11, $p = 0.099$) [L]. PD-L1 < 1% (Proportion): PFS 325 vs 139 days (HR = 1.21, 95% CI 0.86–1.70, $p = 0.266$) [M]; OS 714 vs 383 days (HR = 1.30, 95% CI 0.88–1.93, $p = 0.179$) [N]. PD-L1 ≥ 1% (Proportion): PFS 270 vs 189 days (HR = 1.28, 95% CI 0.92–1.77, $p = 0.139$) [O]; OS NR vs 545 days (HR = 1.27, 95% CI 0.86–1.88, $p = 0.232$) [P]. AD (Proportion): PFS 298 vs 228 days (HR = 1.03, 95% CI 0.75–1.42, $p = 0.855$) [Q]; OS NR vs 591 days (HR = 1.16, 95% CI 0.78–1.71, $p = 0.470$) [R]. non-AD (Proportion): PFS 285 vs 134 days (HR = 1.39, 95% CI 1.00–1.93, $p = 0.046$) [S]; OS 714 vs 291 days (HR = 1.34, 95% CI 0.91–1.97, $p = 0.134$) [T].

Assessment of inflammatory and nutritional indices

Clinical and biological data (total protein, albumin, and C-reactive protein [CRP] levels; white blood cell, neutrophil, platelet, and lymphocyte counts; and height and weight) were extracted from medical records for analysis. Six indices reflecting systemic inflammatory and nutritional status based on previous studies²¹ were calculated at baseline within one week of the first cycle of each treatment. The inflammatory indices used were neutrophil to lymphocyte ratio (NLR), platelet-to-lymphocyte ratio (PLR), and systemic immune-inflammation index (SII) (platelet count $\times$ neutrophil count/lymphocyte count)^{18,22,23}. The nutritional indices included prognostic nutritional index (PNI) = $10 \times$ albumin (g/dL) + $0.005 \times$ lymphocyte count¹⁴; advanced lung cancer inflammation index (ALI) = body mass index + albumin level (g/dL)/NLR^{22,23}, and Glasgow prognostic score (GPS). The optimal cut-off value for the blood markers was the median value, and values greater than this cut-off were defined as high. The GPS was tabulated as follows: 0, no abnormal values (good); 1, abnormal value (intermediate); and 2, two abnormal values (poor)²⁴. Abnormal values were defined as CRP level > 10 mg/mL and albumin < 3.5 g/dL. A GPS of 0 was defined as low, while a GPS of 1 or 2 was classified as high.

Immunohistochemistry (IHC)

The assessment and procedure for immunohistochemical staining were performed as previously described^{4–6}. TROP2 expression was immunohistochemically examined by incubating tumor specimens with a rabbit monoclonal antibody against TROP2 (ab214488, clone EPR20043, high pH retrieval, Abcam, Waltham, MA, USA) at a dilution of 1:2000 in antibody diluent (Dako Glostrup, Denmark; S2022), overnight at 4 °C, followed by incubation at room temperature for 30 min. TROP2 expression was considered positive when its staining was observed in the plasma membrane and cytoplasm. The expression of TROP2 was evaluated by the intensity and proportion scores as described previously^{4–6}. The intensity score was defined as score 0 (negative), score 1 (weak), score 2 (moderate), or score 3 (strong), and the proportion score presented the estimated fraction of positively stained tumor cells (1 = < 10%, 2 = 10–50%, 3 = 50–80%, 4 = 80–100%) (Fig. 1). The immunostaining score of TROP2 was calculated as the product of a proportion score and intensity score. Therefore, the total score ranged from 0–12. Overexpression of TROP2 was defined as a total score of 12. Mouse monoclonal Ki67 (M7240 clone MIB-1, 1:100, pH6 retrieval DAKO, Glostrup, Denmark) was quantified as previously described²⁵. The cutoff value of Ki-67 expression was the median value, and high and low expression levels were defined according to this cutoff.

The sections were evaluated by at least two researchers (ST and KK) using a light microscope ($\times 200$ and $\times 400$ magnification) in a blinded manner. In case of discrepancies, both investigators simultaneously evaluated the slides until a final consensus was reached. The investigators were blinded to the patient outcomes.

Statistical analysis

Continuous variables were dichotomized as described above, and categorical variables were compared using the χ^2 test. The statistical significance level was set at $p < 0.05$. PFS was defined as the time from initial treatment to disease progression or death. Overall survival (OS) was defined as the time from initial treatment to death from any cause. PD-L1 (22C3) expression in tumors was counted as tumor proportional score classified into three groups: < 1%, 1–49%, and $\geq 50\%$. The Kaplan–Meier method was used to estimate survival over time, with survival differences analyzed using the log-rank test. Variables that showed significant differences in the Kaplan–Meier analyses were simultaneously included in the multivariate Cox proportional hazards model. All statistical analyses were performed using GraphPad Prism (v.10.0; GraphPad Software, San Diego, CA, USA) and JMP 16.0 (SAS Institute Inc., Cary, North Carolina, USA).

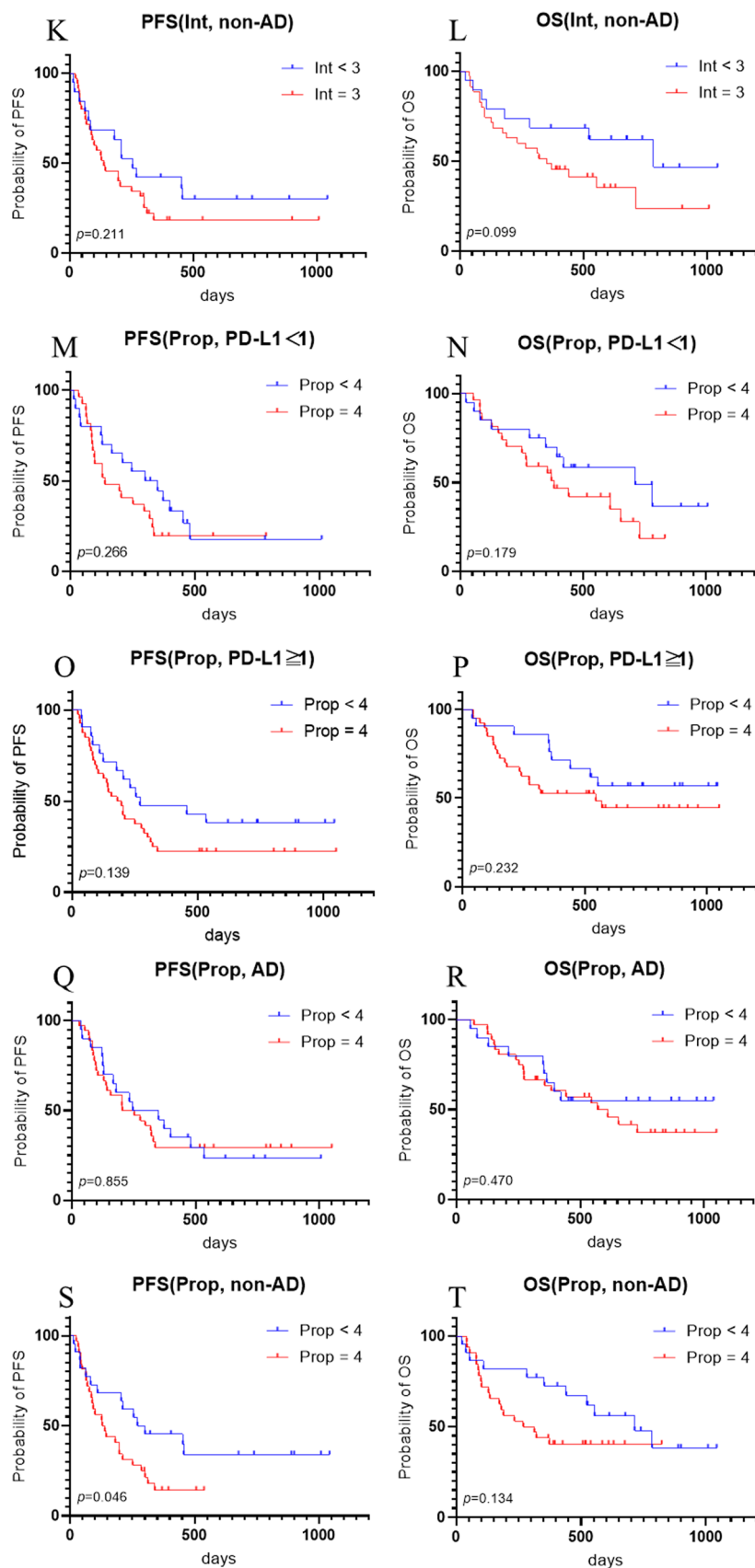


Fig. 3. (continued)

Different variables		Progression-free survival					Overall survival				
		Univariate analysis		Multivariate analysis			Univariate analysis		Multivariate analysis		
		MST (days)	p-value	HR	95% CI	p-value	MST (days)	p-value	HR	95% CI	p-value
Age	< 75/≥ 75 yrs	189/270	0.430				655/443	0.434			
Sex	Male/Female	232/142	0.888				571/395	0.778			
ECOG PS	0–1 / 2–3	270/92	0.001	0.920	0.49–1.73	0.796	784/142	< 0.001	1.377	0.68–2.75	0.368
Smoking (BI)	Yes/≥ No	170/221	0.842				611/395	0.658			
Histology	AC/Non-AC	250/197	0.234				611/523	0.283			
Liver meta	Yes/No	105/251	0.019	2.084	1.10–3.96	0.025	129/714	< 0.001	2.78	1.34–5.40	0.007
Brain meta	Yes/No	232/202	0.919				655/554	0.468			
Bone meta	Yes/No	105/252	0.020	1.787	1.09–2.94	0.023	272/655	0.011	1.92	1.08–3.33	0.027
NLR	High/Low	134/271	0.010	0.427	0.15–1.18	0.100	334/NR	0.004	0.12	0.04–0.42	0.002
PD-L1 (%)	≥ 1/ < 1	205/210	0.611				571/443	0.334			
Ki-67	High/Low	189/208	0.401				611/523	0.834			
TROP2 (Int. × Prop.)	= 12/ < 12	141/273	0.017	2.034	1.27–3.24	0.003	273/784	0.003	2.82	1.59–5.06	< 0.001
PLR	High/Low	138/318	0.001	1.164	0.60–2.26	0.652	283/NR	< 0.001	1.20	0.52–2.51	0.777
SII	High/Low	125/336	< 0.001	3.896	1.56–9.74	0.004	269/NR	< 0.001	5.62	1.87–16.2	0.002
PNI	High/Low	318/126	< 0.001	0.580	0.31–1.08	0.084	NR/233	< 0.001	0.33	0.15–0.73	0.007
ALI	High/Low	318/129	0.002	1.093	0.38–3.13	0.868	NR/262	< 0.001	0.47	0.15–1.69	0.237
GPS	High/Low	200/240	0.466				571/554	0.674			

Table 3. Univariate and multivariate survival analysis in all patients. yrs, years; MST, median survival time (days); ECOG PS, eastern cooperative oncology group performance status; BI, brinkman index; AC, adenocarcinoma; non-AC, non-adenocarcinoma; PD-L1, programmed death ligand-1; BI, brinkman index; lymph, lymph node; HR, hazard ratio; 95% CI, 95% confidence interval; NLR, neutrophil to lymphocyte ratio; PLR, platelet to lymphocyte ratio; SII, systemic immune inflammation index; PNI, prognostic nutrition index; ALI, advanced lung cancer inflammation index; GPS, Glasgow prognostic score; PFS, progression-free survival; OS, overall survival; NR, not reached; bold type, statistically significance; int., intensity; prop., proportion; TROP2, Trophoblast surface antigen 2.

Data availability

All data generated or analyzed during this study are included in this article.

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Author contributions

NH, ST and KK: study conception, design, and manuscript preparation. HI, AM, AS, KH, YM, and OY: patient management. NH, ST, KK, MH, and HI: patient data collection and statistical analyses. NH, KK, ST, MH, and HK: manuscript revision. All authors contributed to and agree with the manuscript content.

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Declarations

Ethical approval

This study was approved by the Institutional Review Board (IRB) of Saitama Medical University, International Medical Center (approval number: 20-125). The requirement for written informed consent was waived by the IRB due to the retrospective study design. In accordance with the IRB-approved protocol, an opt-out method was employed, and patients were given the opportunity to decline participation after public disclosure of study details. All procedures involving human participants were conducted in accordance with the ethical standards of the institutional review board and the 1964 Declaration of Helsinki and its later amendments.

Competing interests

The authors declare no competing interests.

Additional information

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